

Operating Systems

LAB#01



23K-2001

BCS-4J

pwd : display present working directory

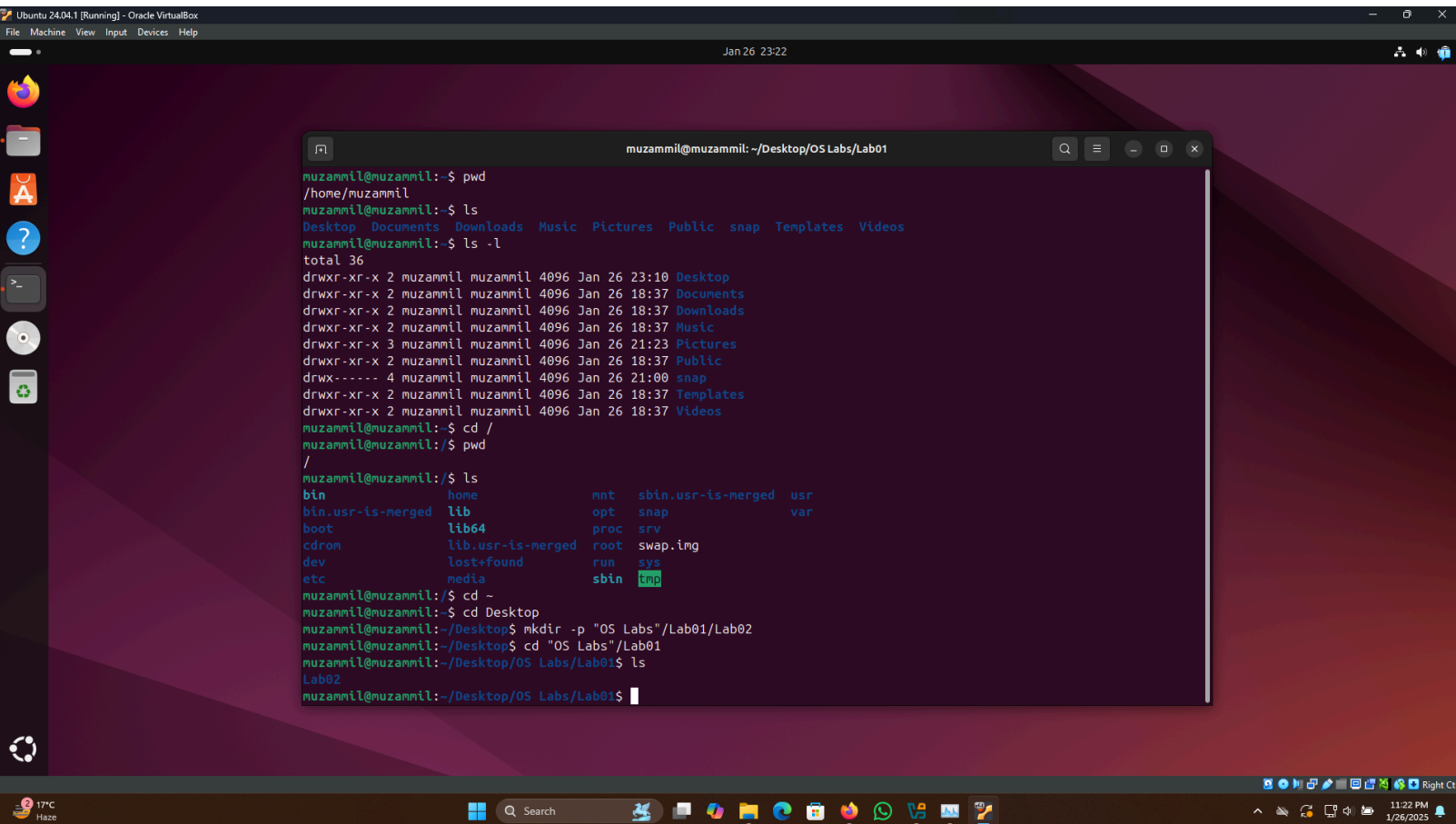
ls : display files/directories

ls -l : display files with details

cd / : change directory to root

cd ~ : change directory to home

mkdir -p parent/child/sub-folder : create directory keeping the first as parent

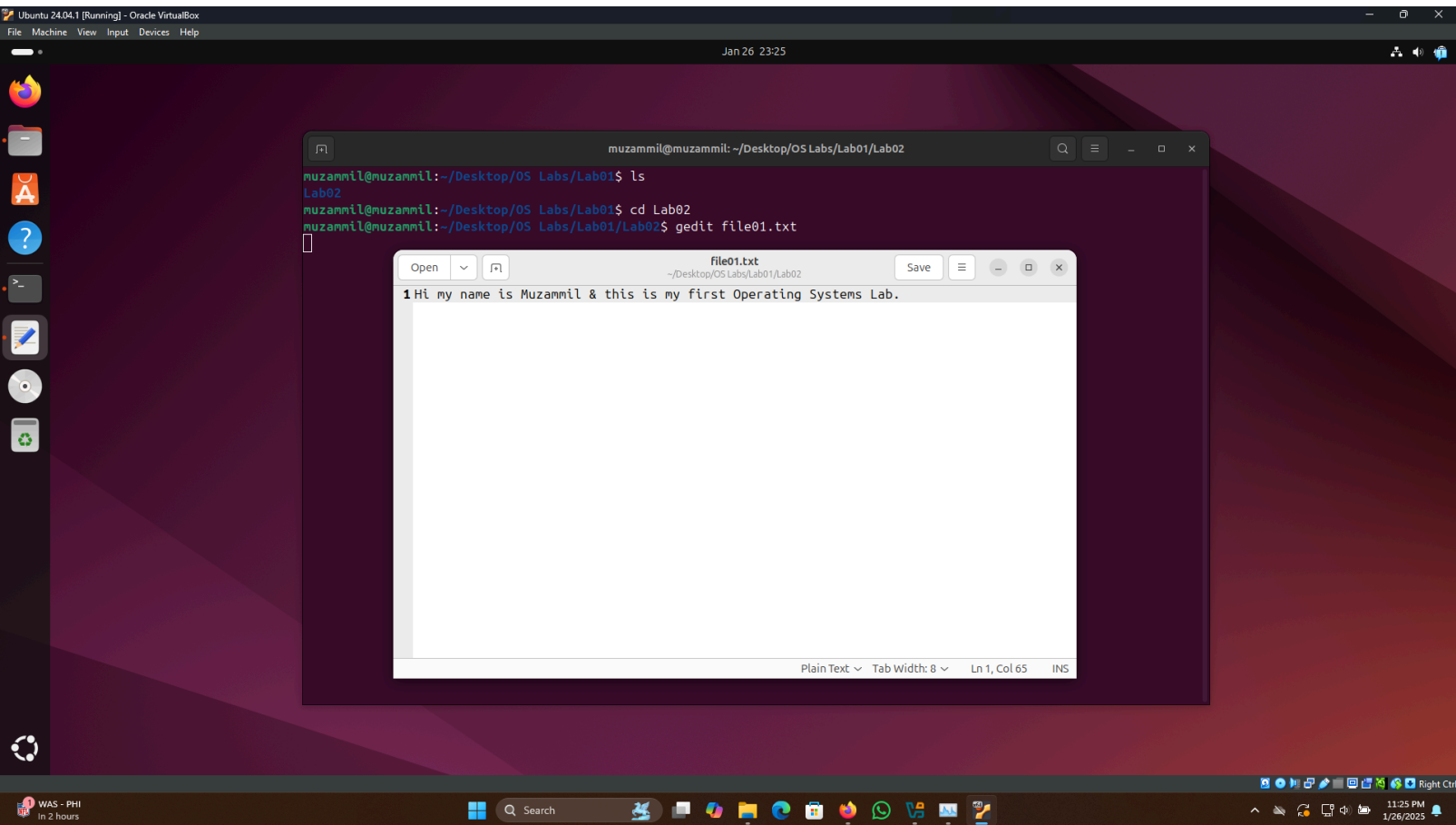


```
Ubuntu 24.04.1 [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
Jan 26 23:22

muzammil@muzammil: ~/Desktop/OS Labs/Lab01
muzammil@muzammil:~$ pwd
/home/muzammil
muzammil@muzammil:~$ ls
Desktop  Documents  Downloads  Music  Pictures  Public  snap  Templates  Videos
muzammil@muzammil:~$ ls -l
total 36
drwxr-xr-x 2 muzammil muzammil 4096 Jan 26 23:10 Desktop
drwxr-xr-x 2 muzammil muzammil 4096 Jan 26 18:37 Documents
drwxr-xr-x 2 muzammil muzammil 4096 Jan 26 18:37 Downloads
drwxr-xr-x 2 muzammil muzammil 4096 Jan 26 18:37 Music
drwxr-xr-x 3 muzammil muzammil 4096 Jan 26 21:23 Pictures
drwxr-xr-x 2 muzammil muzammil 4096 Jan 26 18:37 Public
drwx----- 4 muzammil muzammil 4096 Jan 26 21:00 snap
drwxr-xr-x 2 muzammil muzammil 4096 Jan 26 18:37 Templates
drwxr-xr-x 2 muzammil muzammil 4096 Jan 26 18:37 Videos
muzammil@muzammil:~$ cd /
muzammil@muzammil:/$ pwd
/
muzammil@muzammil:/$ ls
bin             hone            mnt             sbin.usr-is-merged  usr
bin.usr-is-merged  lib             opt             snap              var
boot            lib64           proc            srv
cdrom            lib.usr-is-merged  root            swap.img
dev              lost+found      run             sys
etc              media           sbin            tmp
muzammil@muzammil:/$ cd ~
muzammil@muzammil:~$ cd Desktop
muzammil@muzammil:~/Desktop$ mkdir -p "OS Labs"/Lab01/Lab02
muzammil@muzammil:~/Desktop$ cd "OS Labs"/Lab01
muzammil@muzammil:~/Desktop/OS Labs/Lab01$ ls
Lab02
muzammil@muzammil:~/Desktop/OS Labs/Lab01$
```

gedit filename.txt : open file in text editor (create new if not present)

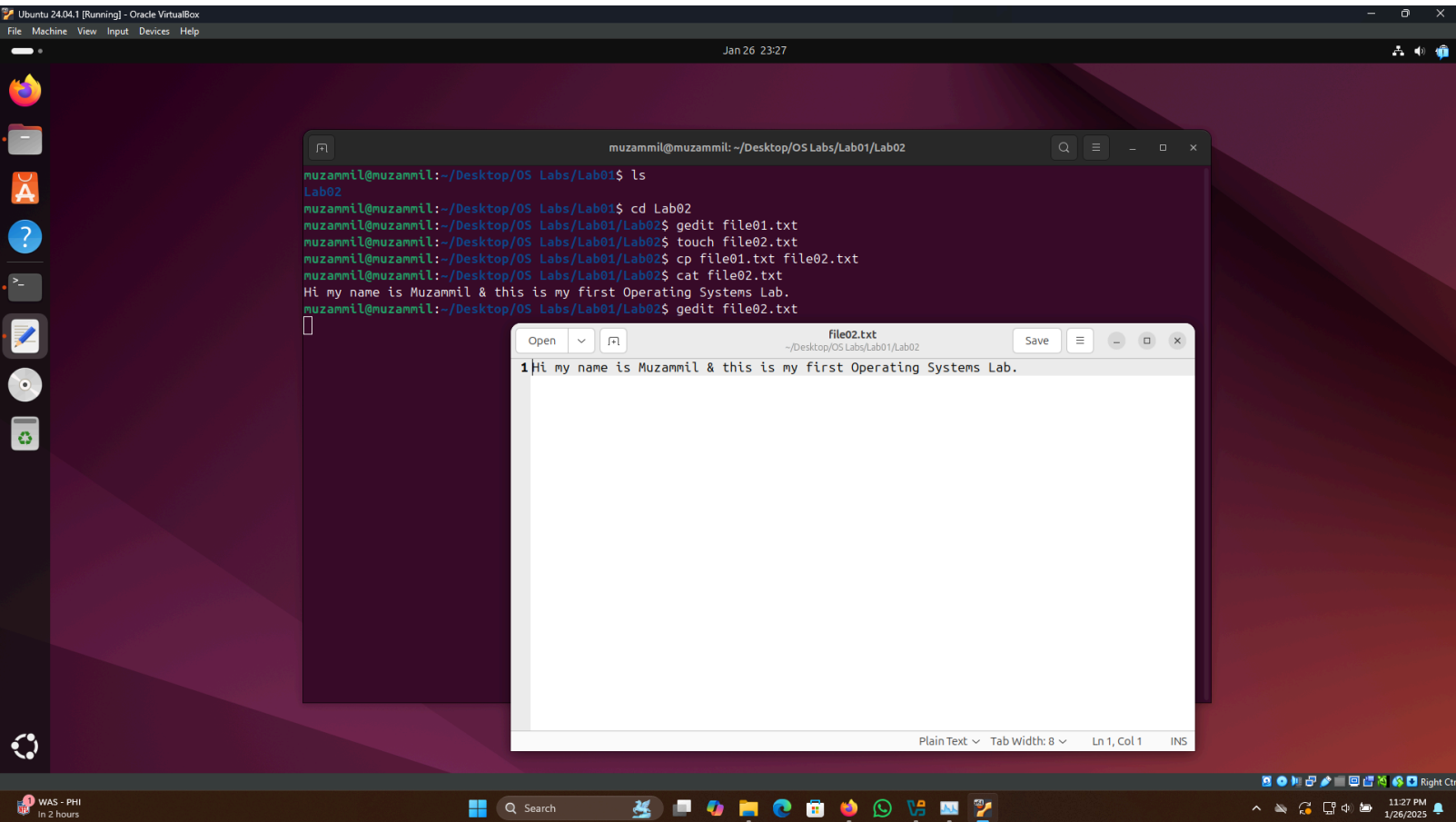
ctrl+z : terminate command



touch filename.txt : create new file

cp file01.txt file02.txt : copy contents of file01.txt to file02.txt

cat file02.txt : display contents of file on terminal



```
muzammil@muzammil: ~/Desktop/OS Labs/Lab01/Lab02
muzammil@muzammil:~/Desktop/OS Labs/Lab01$ ls
Lab02
muzammil@muzammil:~/Desktop/OS Labs/Lab01$ cd Lab02
muzammil@muzammil:~/Desktop/OS Labs/Lab01/Lab02$ gedit file01.txt
muzammil@muzammil:~/Desktop/OS Labs/Lab01/Lab02$ touch file02.txt
muzammil@muzammil:~/Desktop/OS Labs/Lab01/Lab02$ cp file01.txt file02.txt
muzammil@muzammil:~/Desktop/OS Labs/Lab01/Lab02$ cat file02.txt
Hi my name is Muzammil & this is my first Operating Systems Lab.
muzammil@muzammil:~/Desktop/OS Labs/Lab01/Lab02$ gedit file02.txt
```

file02.txt
~ / Desktop / OS Labs / Lab01 / Lab02

1 Hi my name is Muzammil & this is my first Operating Systems Lab.

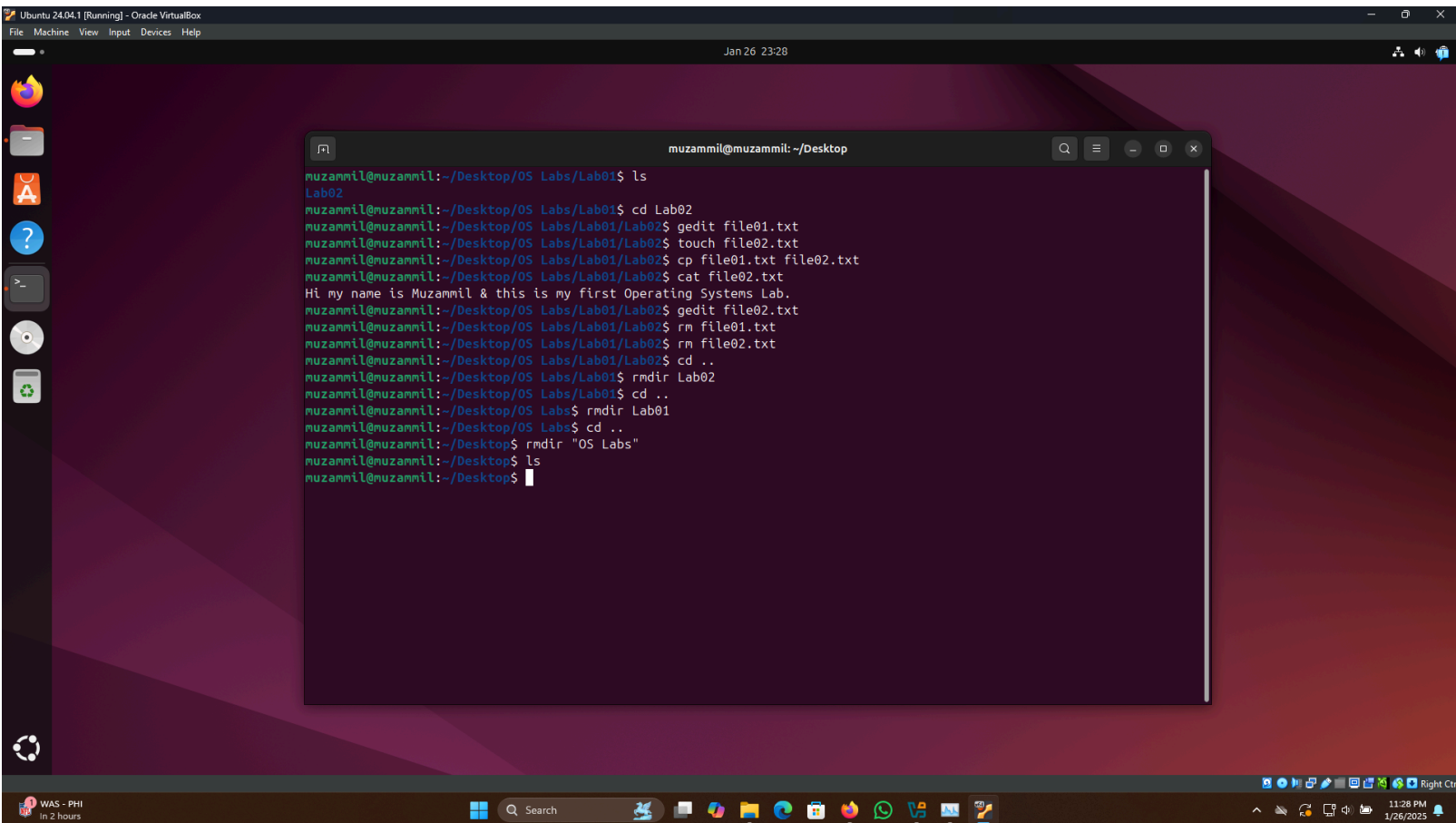
Plain Text ▾ Tab Width: 8 ▾ Ln 1, Col 1 INS

WAS - PHI
In 2 hours

Search

11:27 PM
1/26/2025

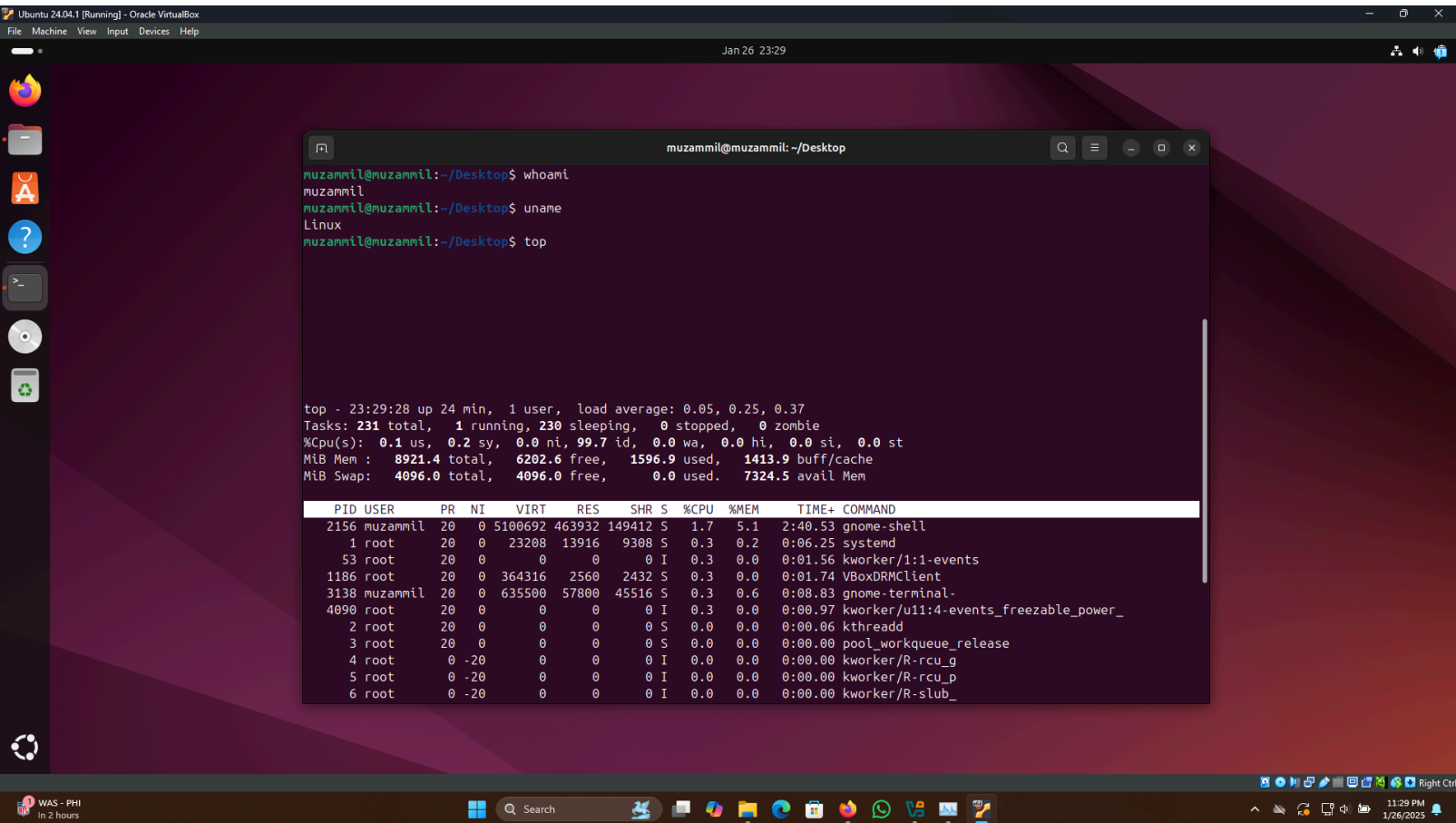
`rm filename.txt` : remove single file
`rmdir folder` : remove single directory (must be empty)



The screenshot shows a terminal window titled 'muzammil@muzammil: ~/Desktop' running on an Ubuntu 24.04.1 virtual machine. The terminal displays a series of commands and their outputs, demonstrating file and directory management. The user navigates to a directory named 'OS Labs/Lab01/Lab02', creates two files ('file01.txt' and 'file02.txt'), copies 'file01.txt' to 'file02.txt', and then removes both files. Finally, the user removes the 'Lab02' directory and returns to the 'OS Labs' directory. The terminal output is as follows:

```
muzammil@muzammil:~/Desktop/OS Labs/Lab01$ ls
Lab02
muzammil@muzammil:~/Desktop/OS Labs/Lab01$ cd Lab02
muzammil@muzammil:~/Desktop/OS Labs/Lab01/Lab02$ gedit file01.txt
muzammil@muzammil:~/Desktop/OS Labs/Lab01/Lab02$ touch file02.txt
muzammil@muzammil:~/Desktop/OS Labs/Lab01/Lab02$ cp file01.txt file02.txt
muzammil@muzammil:~/Desktop/OS Labs/Lab01/Lab02$ cat file02.txt
Hi my name is Muzammil & this is my first Operating Systems Lab.
muzammil@muzammil:~/Desktop/OS Labs/Lab01/Lab02$ gedit file02.txt
muzammil@muzammil:~/Desktop/OS Labs/Lab01/Lab02$ rm file01.txt
muzammil@muzammil:~/Desktop/OS Labs/Lab01/Lab02$ rm file02.txt
muzammil@muzammil:~/Desktop/OS Labs/Lab01/Lab02$ cd ..
muzammil@muzammil:~/Desktop/OS Labs/Lab01$ rmdir Lab02
muzammil@muzammil:~/Desktop/OS Labs/Lab01$ cd ..
muzammil@muzammil:~/Desktop/OS Labs$ rmdir Lab01
muzammil@muzammil:~/Desktop/OS Labs$ cd ..
muzammil@muzammil:~/Desktop$ rmdir "OS Labs"
muzammil@muzammil:~/Desktop$ ls
muzammil@muzammil:~/Desktop$
```

whoami : display username
uname : display operating system name
top : display active process and resources being used



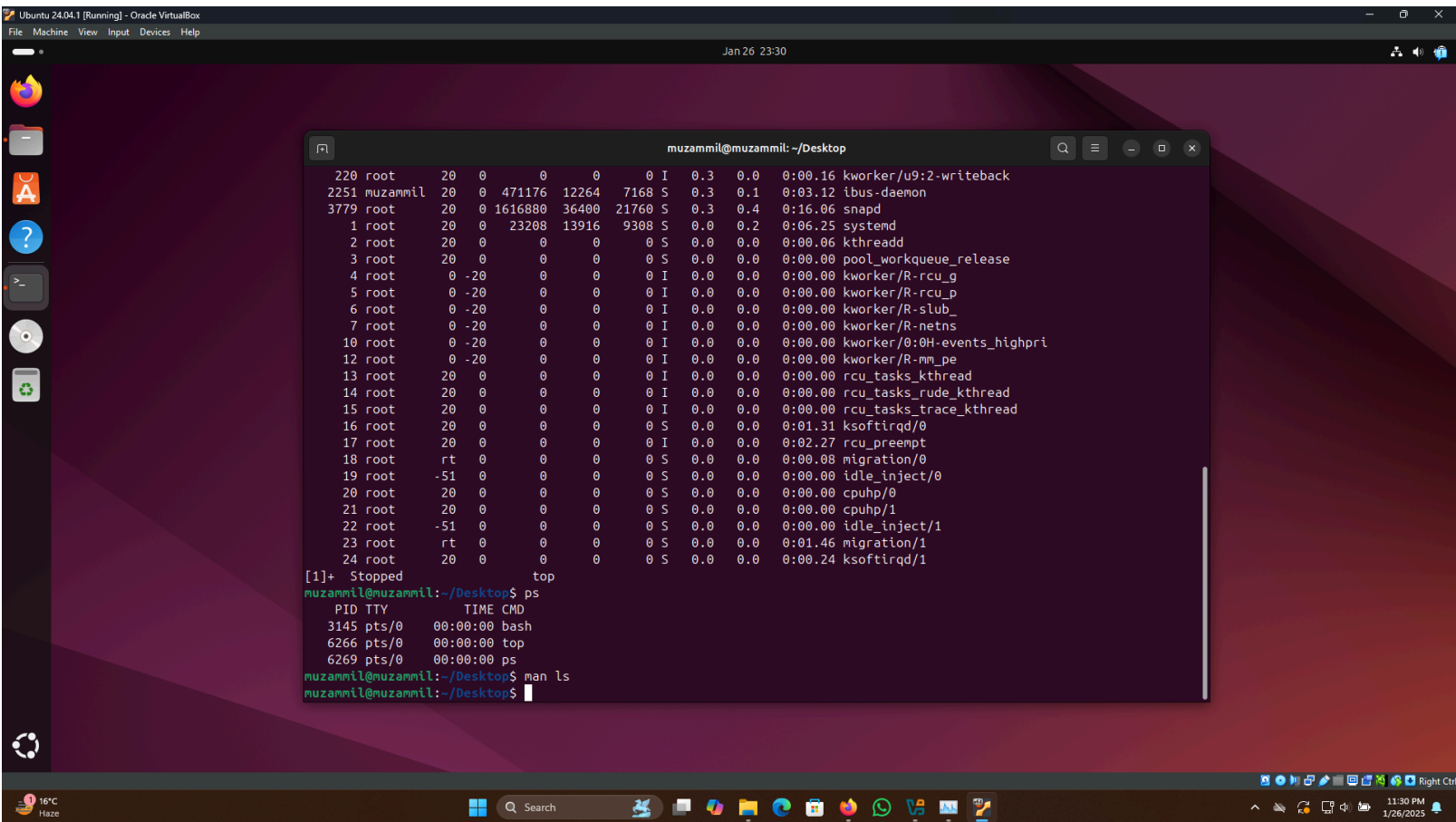
```
Ubuntu 24.04.1 [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
Jan 26 23:29

muzammil@muzammil: ~/Desktop
muzammil@muzammil:~/Desktop$ whoami
muzammil
muzammil@muzammil:~/Desktop$ uname
Linux
muzammil@muzammil:~/Desktop$ top

top - 23:29:28 up 24 min, 1 user, load average: 0.05, 0.25, 0.37
Tasks: 231 total, 1 running, 230 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.1 us, 0.2 sy, 0.0 ni, 99.7 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 8921.4 total, 6202.6 free, 1596.9 used, 1413.9 buff/cache
MiB Swap: 4096.0 total, 4096.0 free, 0.0 used, 7324.5 avail Mem

  PID USER      PR  NI   VIRT   RES   SHR  S  %CPU  %MEM     TIME+ COMMAND
 2156 muzammil  20   0 5100692 463932 149412 S   1.7   5.1   2:40.53 gnome-shell
    1 root      20   0  23208  13916  9308 S   0.3   0.2   0:06.25 systemd
   53 root      20   0     0     0     0 I   0.3   0.0   0:01.56 kworker/1:1-events
 1186 root      20   0 364316  2560  2432 S   0.3   0.0   0:01.74 VBoxDRMClient
 3138 muzammil  20   0 635500  57800 45516 S   0.3   0.6   0:08.83 gnome-terminal-
 4090 root      20   0     0     0     0 I   0.3   0.0   0:00.97 kworker/u11:4-events_freezable_power_
    2 root      20   0     0     0     0 S   0.0   0.0   0:00.06 kthreadd
    3 root      20   0     0     0     0 S   0.0   0.0   0:00.00 pool_workqueue_release
    4 root      0 -20     0     0     0 I   0.0   0.0   0:00.00 kworker/R-rcu_g
    5 root      0 -20     0     0     0 I   0.0   0.0   0:00.00 kworker/R-rcu_p
    6 root      0 -20     0     0     0 I   0.0   0.0   0:00.00 kworker/R-slab_
```

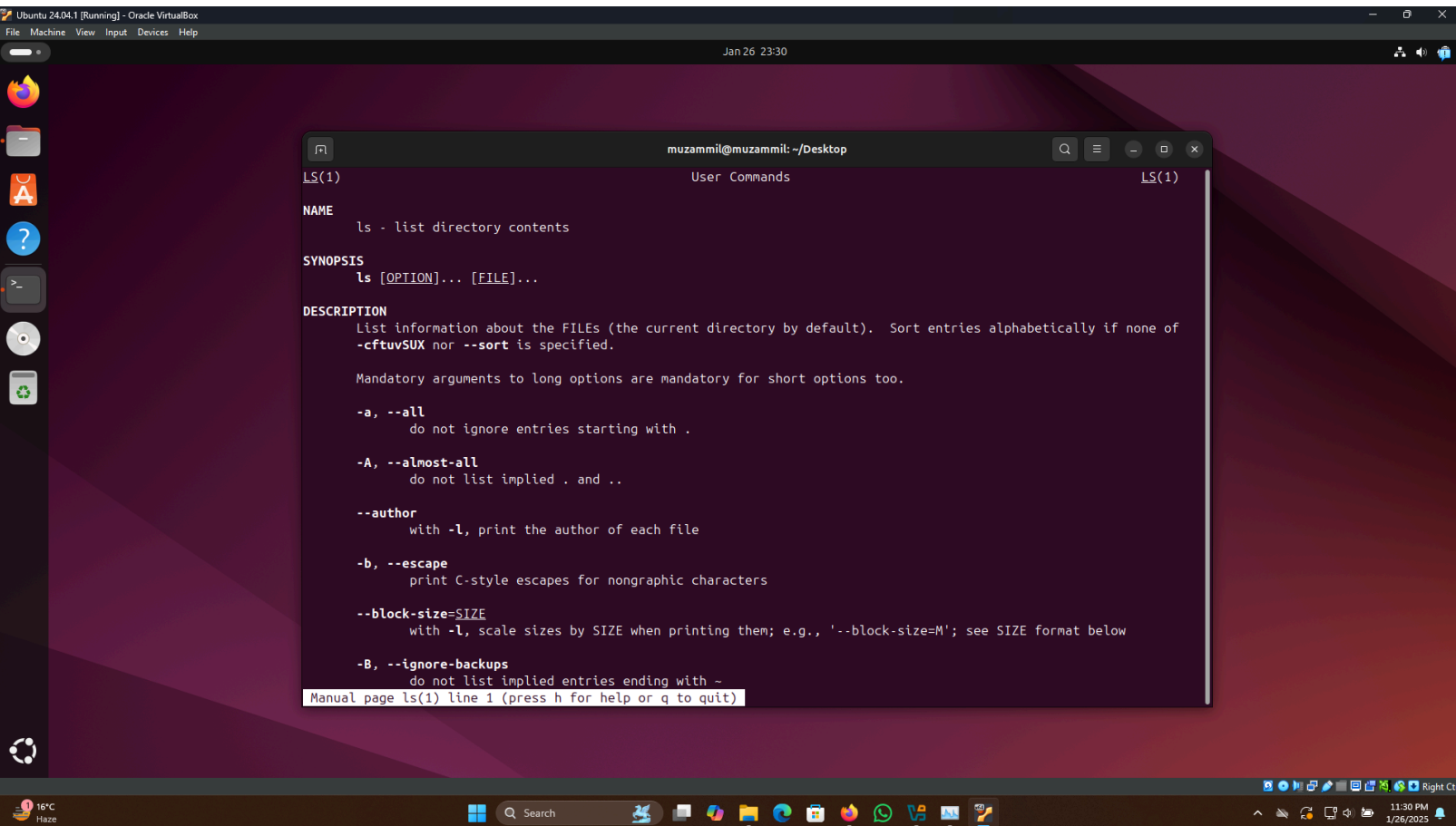
ps : display current processes



The screenshot shows a terminal window titled 'muzammil@muzammil: ~/Desktop' running the 'top' command. The output of 'top' is a list of processes with columns for PID, USER, PR, NI, VIRT, RES, SHR, S, CPU, MEM, and COMMAND. The processes listed include kworker/u9:2-writeback, ibus-daemon, snapd, systemd, kthreadd, pool_workqueue_release, kworker/R-rcu_g, kworker/R-rcu_p, kworker/R-slub, kworker/R-netns, kworker/0:0H-events_highpri, kworker/R-mm_pe, rcu_tasks_kthread, rcu_tasks_rude_kthread, rcu_tasks_trace_kthread, ksoftirqd/0, rcu_preempt, migration/0, idle_inject/0, cpuhp/0, cpuhp/1, idle_inject/1, migration/1, and ksoftirqd/1. Below the 'top' output, the user has entered 'ps' and the output shows the current processes: bash, top, and ps. The user has also entered 'man ls' and 'ps' again.

```
muzammil@muzammil: ~/Desktop
220 root      20   0   0   0   0 I   0.3   0.0   0:00.16 kworker/u9:2-writeback
2251 muzammil  20   0 471176 12264 7168 S   0.3   0.1   0:03.12 ibus-daemon
3779 root      20   0 1616880 36400 21760 S   0.3   0.4   0:16.06 snapd
1 root       20   0 23208 13916 9308 S   0.0   0.2   0:06.25 systemd
2 root       20   0   0   0   0 S   0.0   0.0   0:00.06 kthreadd
3 root       20   0   0   0   0 S   0.0   0.0   0:00.00 pool_workqueue_release
4 root       0 -20   0   0   0 I   0.0   0.0   0:00.00 kworker/R-rcu_g
5 root       0 -20   0   0   0 I   0.0   0.0   0:00.00 kworker/R-rcu_p
6 root       0 -20   0   0   0 I   0.0   0.0   0:00.00 kworker/R-slub
7 root       0 -20   0   0   0 I   0.0   0.0   0:00.00 kworker/R-netns
10 root      0 -20   0   0   0 I   0.0   0.0   0:00.00 kworker/0:0H-events_highpri
12 root      0 -20   0   0   0 I   0.0   0.0   0:00.00 kworker/R-mm_pe
13 root       20   0   0   0   0 I   0.0   0.0   0:00.00 rcu_tasks_kthread
14 root       20   0   0   0   0 I   0.0   0.0   0:00.00 rcu_tasks_rude_kthread
15 root       20   0   0   0   0 I   0.0   0.0   0:00.00 rcu_tasks_trace_kthread
16 root       20   0   0   0   0 S   0.0   0.0   0:01.31 ksoftirqd/0
17 root       20   0   0   0   0 I   0.0   0.0   0:02.27 rcu_preempt
18 root       rt   0   0   0   0 S   0.0   0.0   0:00.00 migration/0
19 root      -51   0   0   0   0 S   0.0   0.0   0:00.00 idle_inject/0
20 root       20   0   0   0   0 S   0.0   0.0   0:00.00 cpuhp/0
21 root       20   0   0   0   0 S   0.0   0.0   0:00.00 cpuhp/1
22 root      -51   0   0   0   0 S   0.0   0.0   0:00.00 idle_inject/1
23 root       rt   0   0   0   0 S   0.0   0.0   0:01.46 migration/1
24 root       20   0   0   0   0 S   0.0   0.0   0:00.24 ksoftirqd/1
[1]+  Stopped                  top
muzammil@muzammil:~/Desktop$ ps
  PID TTY          TIME CMD
 3145 pts/0    00:00:00 bash
 6266 pts/0    00:00:00 top
 6269 pts/0    00:00:00 ps
muzammil@muzammil:~/Desktop$ man ls
muzammil@muzammil:~/Desktop$
```

man ls : list the manual



The screenshot shows a terminal window titled "muzammil@muzammil: ~/Desktop" with a search bar and window controls. The terminal displays the manual for the 'ls' command, which is titled "LS(1) User Commands LS(1)". The manual is organized into sections: NAME, SYNOPSIS, and DESCRIPTION. The NAME section states "ls - list directory contents". The SYNOPSIS section shows "ls [OPTION]... [FILE]...". The DESCRIPTION section explains that 'ls' lists information about files, sorts them alphabetically by default, and lists mandatory arguments for long options. It then lists several options: -a, --all (do not ignore entries starting with .), -A, --almost-all (do not list implied . and ..), --author (with -l, print the author of each file), -b, --escape (print C-style escapes for nongraphic characters), --block-size=SIZE (with -l, scale sizes by SIZE when printing them; e.g., '--block-size=M'; see SIZE format below), and -B, --ignore-backups (do not list implied entries ending with ~). The terminal window is open on a desktop environment with a dark purple background and a taskbar at the bottom showing various application icons and system status information like temperature and time.

```
LS(1) User Commands LS(1)

NAME
  ls - list directory contents

SYNOPSIS
  ls [OPTION]... [FILE]...

DESCRIPTION
  List information about the FILES (the current directory by default). Sort entries alphabetically if none of
  -cftuvSUX nor --sort is specified.

  Mandatory arguments to long options are mandatory for short options too.

  -a, --all
      do not ignore entries starting with .

  -A, --almost-all
      do not list implied . and ..

  --author
      with -l, print the author of each file

  -b, --escape
      print C-style escapes for nongraphic characters

  --block-size=SIZE
      with -l, scale sizes by SIZE when printing them; e.g., '--block-size=M'; see SIZE format below

  -B, --ignore-backups
      do not list implied entries ending with ~
Manual page ls(1) line 1 (press h for help or q to quit)
```